Fundamentals of Data Science(22CSC59)

BE- CSE-V-Semester

Presented by
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Unit-II

Topics to be covered

Techniques: Introduction: Data Analysis and Data Analytics

Descriptive Analysis: Variables, Frequency Distribution, Measures of Centrality, Dispersion of a Distribution.

Diagnostic Analytics: Correlations, Predictive Analytics, Prescriptive Analytics, Exploratory Analysis, Mechanistic Analysis: Regression.

Statistics: Understanding attributes and their types, Types of attributes, Discrete and continuous attributes, measuring central tendency: Mean, Mode, Median, measuring dispersion, Skewness and kurtosis, understanding relationships using covariance and correlation coefficients: Pearson's correlation coefficient, Spearman's rank correlation coefficient, collecting samples, performing parametric tests.

Descriptive Statistics: Understanding statistics, distribution functions uniform distribution, normal distribution, exponential distribution, binomial distribution. Cumulative distribution function, descriptive statistics.

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Data analysis and Data analytics

- **Data Analysis** is the process of inspecting, cleaning, transforming, and interpreting data to discover meaningful patterns, trends, relationships, and insights that help answer specific questions or test hypotheses.
- Data analytics is the systematic science of collecting, cleaning, and examining raw data to discover meaningful patterns, predict future outcomes, and drive smarter decisions.
- While raw data is a massive collection of unorganized numbers and facts, data analytics builds the processes, algorithms, and visualization tools that turn that noise into actionable strategy.

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Example:

building a house: data analytics

inspecting a quality of the house: data analysis.

- Data analysis uses historical (past) data to determine what happened.
- Data Analytics: Uses historical and current data, along with statistical and machine learning techniques, to answer "Why did it happen?", "What will happen?", and "What should we do?"

Data analysis	Data analytics
Analyse historical data to understand what happened	uses statistical methods and machine learning models to generate insights and supports the future decision
Understands the past events	Predict the future outcomes and recommendation actions
Reports, charts, dashboards, summaries.	Predictions, forecasts, recommendations, decision support.

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There are six classes of analysis and analytic techniques

1. **Descriptive analysis-** what happened?
2. **Diagnostic analytics-** why did it happen?
3. **Predictive Analytics** - What will happen?
4. **Prescriptive Analytics-** What should we do?
5. **Exploratory Analysis** - What patterns exist?
6. **Mechanistic Analysis** - What causes the event?

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1. Descriptive analysis

- Also known as descriptive analytics.
- The process of summarizing and organizing data to understand what has happened.
- to analyze historical data to make data-driven decisions, improve business processes, and extract meaningful insights for future planning.

Key points about descriptive analysis

1. First step of data analysis on dataset.
2. Applied to larger datasets such as census or sales data
3. Description(summarized) and interpretation(draws conclusions)
4. Descriptive analysis helps to identify patterns, trends and distribution of data.
5. Uses descriptive statistics to summarize the data effectively.

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1.1 variable

- Numerical representation is more precise than words
- Dataset values to be represented in numerical values.
- Numbers describes the objects more accurately than words.
- Variable is a label or attributes that stores information about object.
- Column in a dataset is a variable.

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Real World	Numerical Representation
Small room	Room size = 100 sq. ft.
State population	18,801,318 people
Student performance	SAT Score = 800

Types of variables

1. Categorical variable
2. Nominal variable
3. Ordinal variable
4. Internal variable
5. Ratio variable
6. Independent variable/ predictor variable
7. Dependent variable/ outcome variable
8. Predictor variable
9. Outcome variable

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1. **Categorical variable:** whose values are categories or groups.

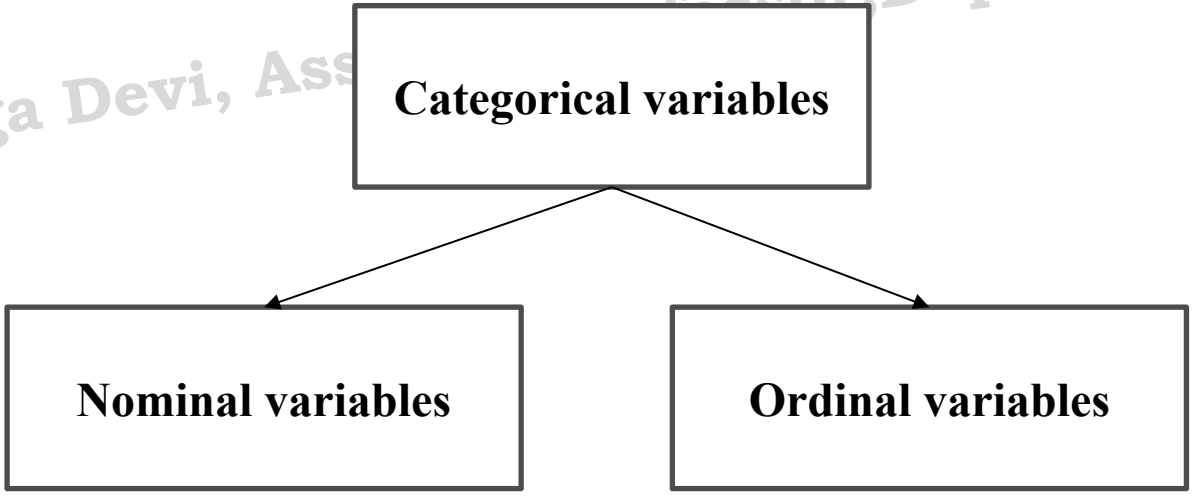
examples: Gender

department

blood group

colors

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2. Nominal variables: there is no order among the categories

- examples: Gender, blood group, City.
  - there is no ranking in between blood group (A+,B+,O+).

3. Ordinal variables: variables whose values can be ordered or ranking, but differences between them not measurable

- example: Excellent>Good>Average>Poor

  Child<Teenage<Adult<old age

4. Internal variables: a numerical variable where differences are meaningful.

- Example : temperature today- 30°C
- temperature tomorrow- 40°C - tomorrow temperature likely to increased by 10 degrees.
- Multiplication and divisions not allowed here.
- Ex: tomorrow's temperature is double todays temperature.
- Its value should not be true Zero

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5. Ratio variables: numerical variable that has equal intervals and true zero.

- can perform all the arithmetic operations which has meaning.

- Examples: height, weight, age, kgs etc.

Day	Orders
Monday	100
Tuesday	200
Wednesday	300

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- Tuesday received twice as many orders as Monday

- Wednesday received thrice as many orders as Monday

- no of orders is Ratio variables.

6. Independent variables/ Predictor variable: A variable whose values will not affect

by other variables . Also called as predictor variable.

Example: house price prediction

area, bedrooms, location- Independent variables

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7. Dependent variables/ outcome variable: values depends on one or more independent variables.

- also called as outcome variable

Example: house price prediction- dependent variable

area, bedrooms, location- Independent variables

Variable Type	Description	Example
Categorical	Represents categories	Gender, Branch
Nominal	Categories without order	Blood Group
Ordinal	Categories with order	Satisfaction Level
Interval	Numeric, no true zero	Temperature (°C)
Ratio	Numeric with true zero	Height, Weight, Age
Independent	Predictor/Input variable	Area, Tumor Size
Dependent	Outcome/Target variable	House Price, Cancer

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1.2. Frequency distribution

- Before analysis we must know what are the variables available.
- We must know how many times each value appears in dataset.
- How the data is distributed across the dataset?
- Frequency distribution is organizing the data and determining how often each value occurs.
- Shows how many time each value appears in dataset
- Distribution of data presented using structured table or graph
- Various methods to represent the frequency distribution

1. Histogram 2. Bar graphs 3. pie charts.

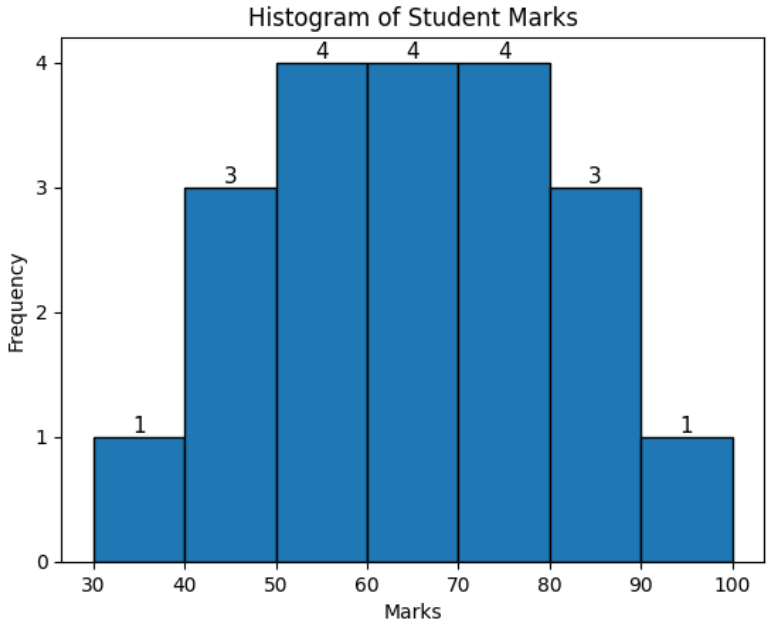
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Histogram

- A histogram is a graphical chart that shows how numerical data is distributed across continuous intervals, or "bins," using connected vertical bars. The horizontal axis displays the ranges of values or bins, while the vertical axis shows the frequency or count of data points in each range.
- Example

marks = [35, 42, 45, 48, 52, 55, 57, 58, 60, 62, 65, 67, 70, 72, 75, 78, 80, 82, 85, 90]

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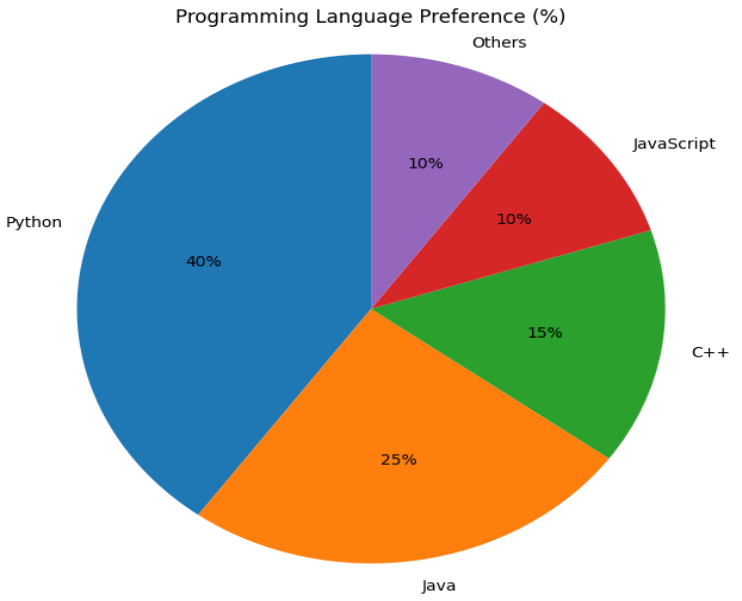


Pie charts

Pie chart is use for the categorical data

Programming Language	Students	Percentage
Python	40	40%
Java	25	25%
C++	15	15%
JavaScript	10	10%
Others	10	10%

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- Above pie chart displays the percentage of each category.
- Percentage distribution of students' preferred programming languages.

There are six classes of analysis and analytic techniques

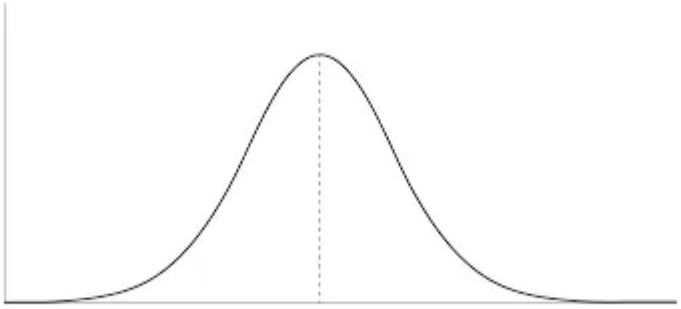
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Normal distribution

- A normal distribution is a symmetric, bell-shaped distribution where data is evenly distributed around the center (mean). If a vertical line is drawn through the center, both sides of the distribution are mirror images of each other.
- Normal distribution is also called Gaussian Distribution.

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Source: Google

Characteristics:

- **bell- shaped curve**
- **symmetric about the centre**
- **Mean, median and mode are equal**
- **Most observations are close to centre**
- **Some observations are extreme**

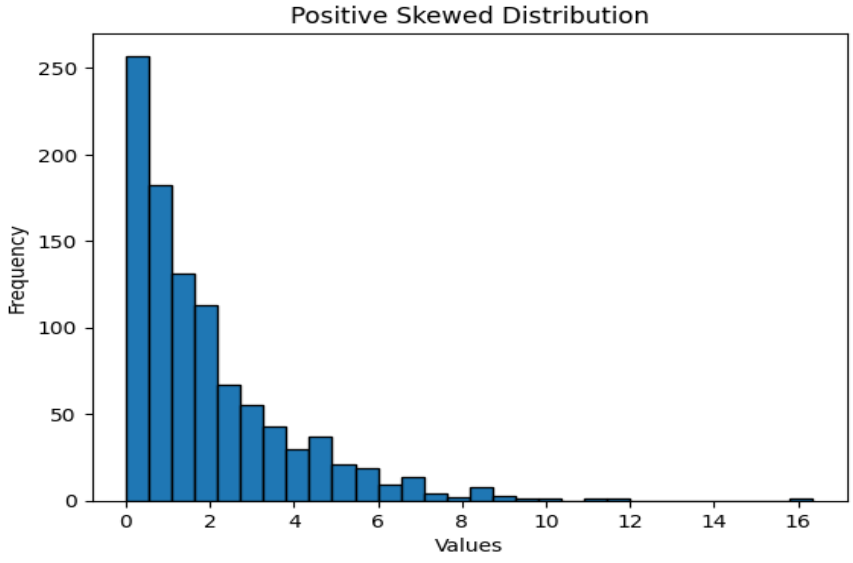
- Distribution of data values can differ from normal distribution in two ways

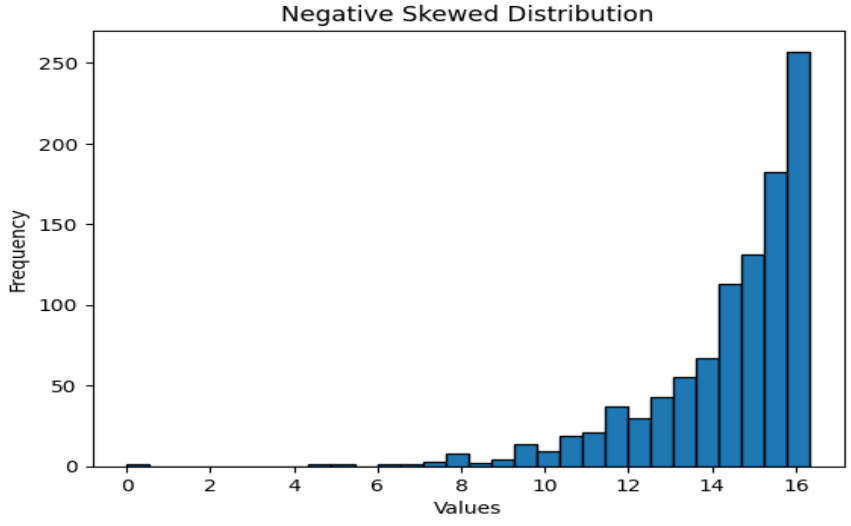
1. skewness: data distribution is not symmetric (one side is longer than the other side)

    i. positively skewed distribution – distributed more on right side

    ii. Negatively skewed distribution.- distributed more on left side.

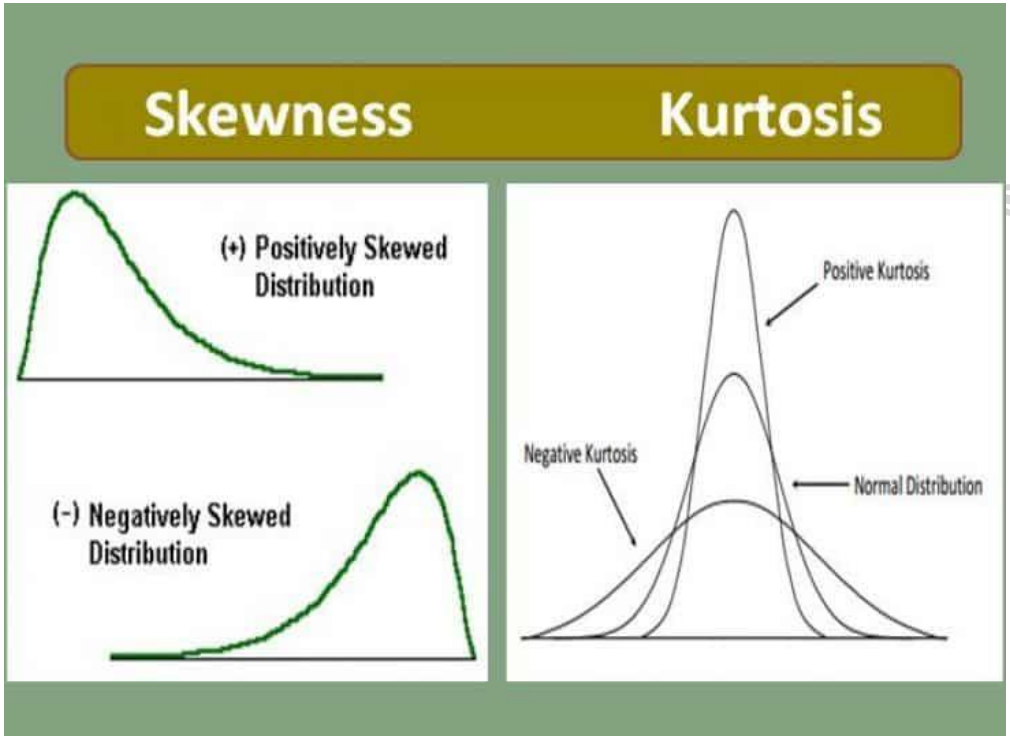
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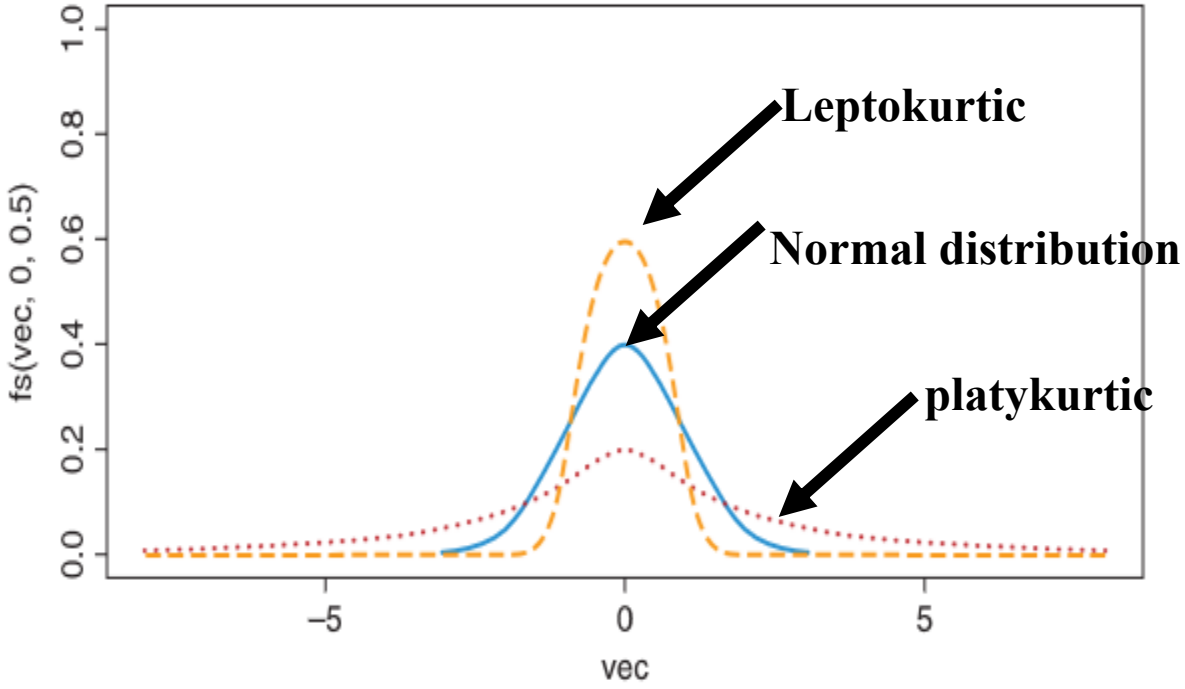
2. kurtosis: distribution of data is either more peaked or flatter than a normal distribution.

Below figure shows skewness and kurtosis.



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https://www.excelr.com/blog/techonologies/skewness-and-kurtosis



Examples of different kurtosis in a distribution (orange dashed line represents leptokurtic, blue solid line represents the normal distribution, and red dotted line represents platykurtic).

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1.3 Measures of centrality

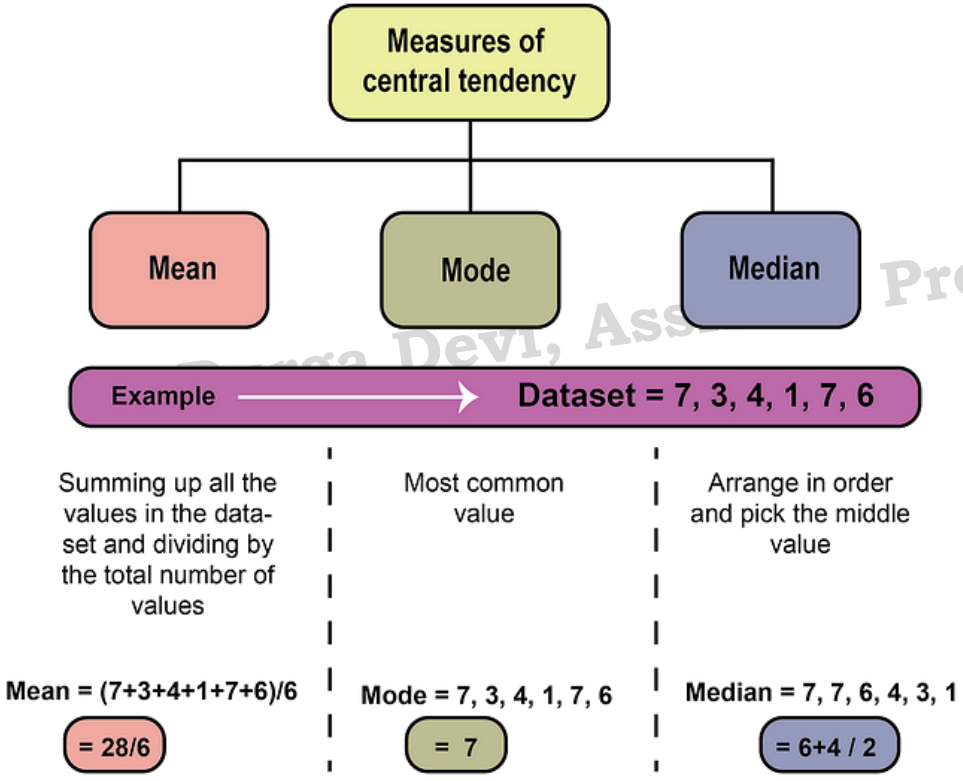
- Also called measure of central tendency.
- Answers what is typical or central value of the whole dataset.
- It attempt to describe the whole set of data with a single value that represents the middle of the dataset.
- Example : student marks={ 90,80,70,98,87,90,68…….}

  - average marks of the class is 80 .

  - 80 represents the measure of centrality.

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There are three main measures of central tendency:
1. Mean 2. Mode 3. Median



Mean	Median	Mode
The Average	The middle value	Most frequent value
Sum all values%N	Sort data and pick middle value	Count occurrences and find max value

1. Mean

1. Mean is known as average

- **If there are n number of values in a dataset and the values are $x_1, x_2, x_3, \ldots, x_n$, then the mean is calculated as**

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

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- Mean is suitable if dataset is normally distributed.
- Mean is highly affected by extreme values(outliers) that leads to skewed distribution.
- To overcome this median to be used.

2. Median

- Median is the middle value of the dataset.
- Sort all the values and take the middle value as a median.
- For example, we have data below as:

- 65, 55, 89, 56, 35, 14, 56, 55, 87, 45. what is median of these values?

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- 14, 35, 45, 55, 55, 56, 56, 65, 87, 89.
- Median is $55 + 56 \div 2 = 111 \div 2 = 55.5$.

Mean and median

65, 55, 89, 56, 35, 14, 56, 55, 87, 45, 92, **500**

Here outlier is 500.

Measure	Before Adding Outlier	After Adding Outlier
Mean	**59**	**95.75**
Median	**56**	**56**

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Drawback: mean and median used for numerical values not for the categorical values.

3. Mode

Mode: most frequently occurring value in the dataset or distribution.

Example:

Consider this dataset showing the retirement age of 11 people, in whole years:
54, 54, 54, 55, 56, 57, 57, 58, 58, 60, 60.

This table shows a simple frequency distribution of the retirement age data.

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Age	Frequency
54	3
55	1
56	1
57	2
58	2
60	2

Mode is 54.

The mode has an advantage over the median and the mean as it can be found for both numerical and categorical (non-numerical) data.

1.4 Dispersion of distribution

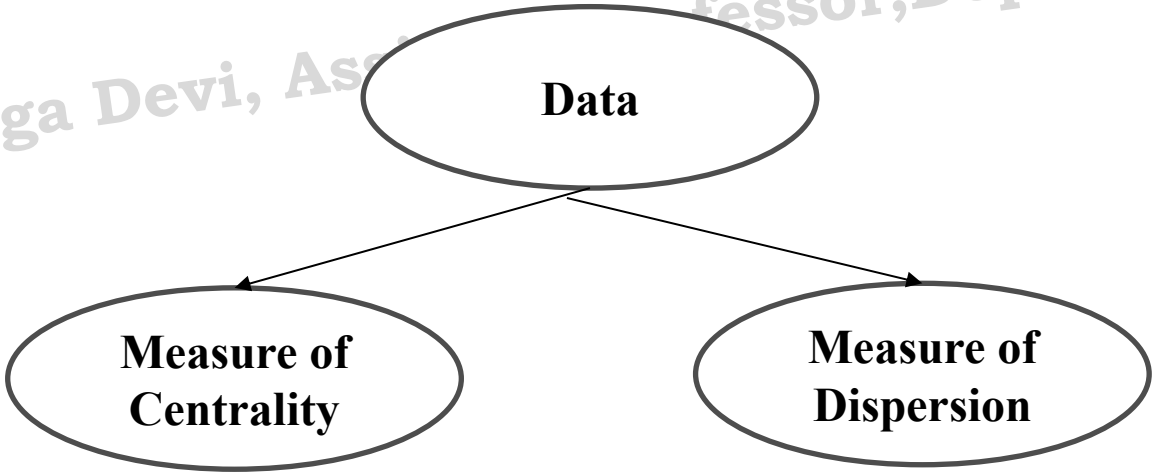
What is dispersion?

Class A: 78, 79, 80, 81, 82 - mean =80

Class B: 40, 60, 80, 100, 120 - mean=80

Dispersion is measure how data is spread out around the typical value?

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Types of measure of dispersion

Measure of dispersion is degree to which data is spread out or scattered around an average or typical value.

1. Range
2. Interquartile Range(IR)
3. Variance
4. Standard deviation

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Measure of dispersion helps to

- Understand variability of the data
- Compare consistency of two datasets
- Detect the outliers

1. Range

Range is simplest method of measuring dispersion of data.

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

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- Example
- Student marks= 20, 35, 40, 60.
- Range=60-20=40

Dataset A
20, 30, 40, 50, 60
Range = 60 − 20 = **40**
Dataset B
20, 21, 22, 23, 60
Range = 60 − 20 = **40**

- Drawback : will be affected by outliers.

2. Interquartile Range

- Range method is affected by the outliers (extremely large or small values).
- IQR measures the spread of the middle 50% of the data.
- It ignores the extreme values.
- Focus on central data.
- Gives the clear view of how the data values spread around the median.

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Procedure:

1. Sort the dataset
2. Q1(lower quartile)- take lower 25% of the data
3. Q2(second quartile)- take median of the whole dataset
4. Q3(third quartile)- higher 25% of the data.
5. **IQR=Q3-Q1**

Example on IQR

- Examples

marks = [20, 25, 30, 35, 40, 45, 50, 55, 60]

Q1=[20,25,30,35]= (25+30)/2=27.5

Q2= median=40

Q3=[45,50,55,60]=(50+55)/2=52.5

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IQR=Q3-Q1=25 . So, the middle 50% of students scored within a range of 25 marks.

Lower half(Q1=27.5)	Median(Q2)	Upper half(Q3=52.5)
20,25,30,35	40	45, 50, 55, 60

3. variance

- Measure how far each data point from centre (mean of the data).
- Helps to understand the degree of variability or consistency withing the data set.
- Uses every observation.

$$\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{N}$$

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Where

x^i is ith value

$\bar{x}$ is mean

- This formula calculates the population variance. It measures the average squared distance of all data values from the mean.

Example on variance

Find the Population Variance of the Data 5, 7, 9, 10, 14, 15.

1. Calculate mean

Mean ($\bar{x}$)=5+7+9+10+14+15=60/6=10

Data(x)	$(x - \bar{x})^2$	Value of $(x - \bar{x})^2$
5	$(5 - 10)^2$	25
7	$(7 - 10)^2$	9
9	$(9 - 10)^2$	1
10	$(10 - 10)^2$	0
14	$(14 - 10)^2$	16
15	$(15 - 10)^2$	25

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Drawback

Variance presents squared units rather than original units.

- Difficult to interpret

Variance = (25 + 9 + 1 + 0 + 16 + 25) / 6 = 76/6 = 12.67

4. Standard deviation

- Variance is expressed in squared units, making it difficult to interpret.
- Suppose student marks in squared units is not meaningful unit.
- So square root the variance is called standard deviation.

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

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- a statistical measure that shows how spread out numbers are from their average (mean), using the exact same units as the data.

Standard deviation



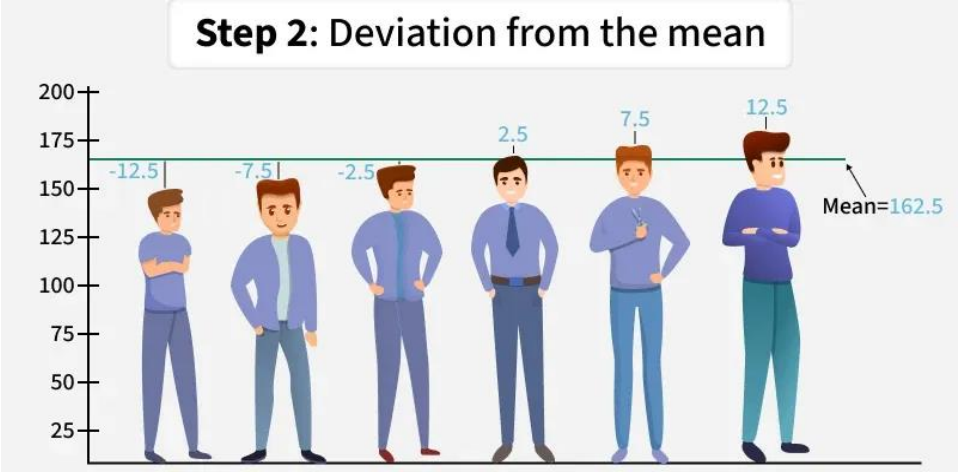
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Step-1 calculate the mean

162.5 - average height

Step-2: deviation from mean

datavalue- mean



Standard deviation

Step-3 calculate the deviation(variance)

Step 3: Square the deviation(Variance)

$$\sigma^2 = \frac{(-12.5)^2+(-7.5)^2+(-2.5)^2+(12.5)^2+(7.5)^2+(2.5)^2}{6}$$

$$\sigma^2 = \frac{156.25+56.25+6.25+156.25+56.25+6.25}{6}$$

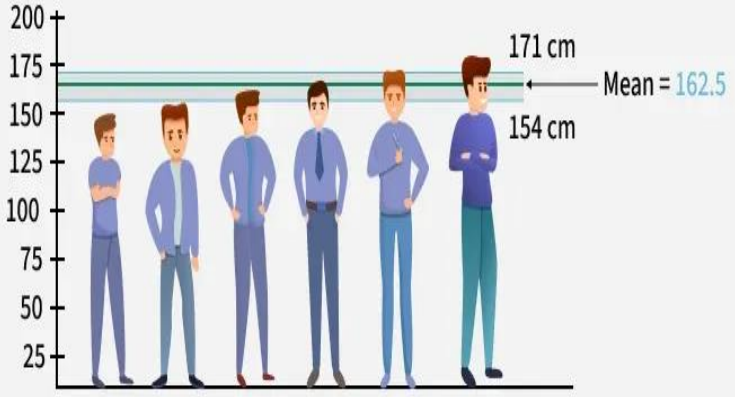
$$\sigma^2 = \frac{437.5}{6}$$

$$\sigma^2 = 72.92$$

Step-4: calculate standard deviation

Step 4: Calculate the Standard Deviation

Square root of Variance = $\sqrt{72.92}$ = 8.5 cm



Standard Deviation shows the typical distance from the mean

On average, each person's height is about 8.5 cm away from the average height (162.5 cm).

https://www.geeksforgeeks.org/maths/standard-deviation-formula/

4 TYPES OF ANALYTICS – DOCTOR & PATIENT EXAMPLE

From understanding the problem to taking the best action

1. DESCRIPTIVE

What happened?



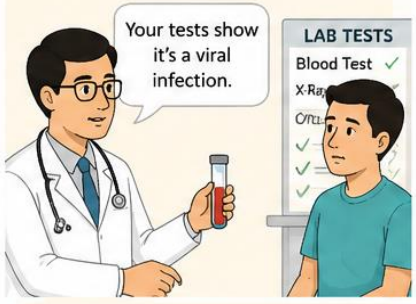
The doctor records and summarizes the patient's symptoms and signs.

Example:
Fever: 102°F, Cough, Headache, Tiredness

Focus:
Understanding what is happening.

2. DIAGNOSTIC

Why did it happen?



The doctor analyzes test results and medical history to find the root cause.

Example:
Viral infection is the cause of the fever.

Focus:
Finding the cause of the problem.

3. PREDICTIVE

What will happen?



The doctor uses data and past patterns to predict the possible outcomes.

Example:
You may recover in 5–7 days with treatment. Without treatment, the condition may worsen.

Focus:
Forecasting future outcomes.

4. PRESCRIPTIVE

What should we do?



The doctor recommends the best actions to achieve the desired outcome.

Example:
Take medicines, rest, drink fluids and come for follow-up.

Focus:
Recommending the best action to achieve the goal.

In short: Understand the problem (Descriptive) → Find the cause (Diagnostic) → Predict the future (Predictive) → Take the best action (Prescriptive)

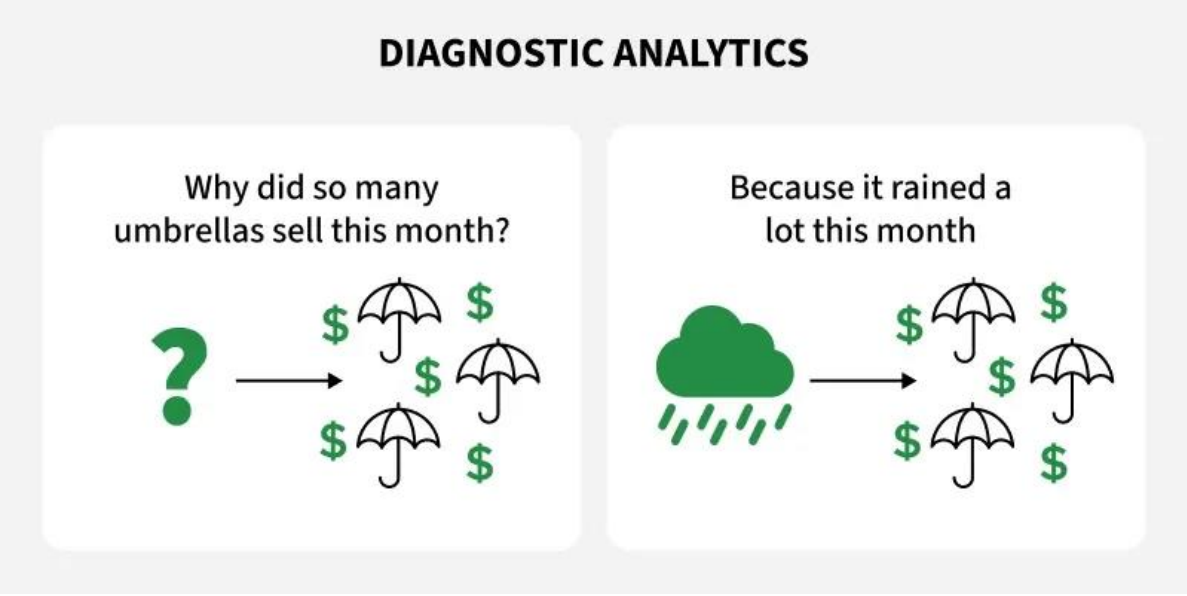
Data + Analysis + Decision = Better Health Outcomes

Generated by AI

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2. Diagnostic analytics

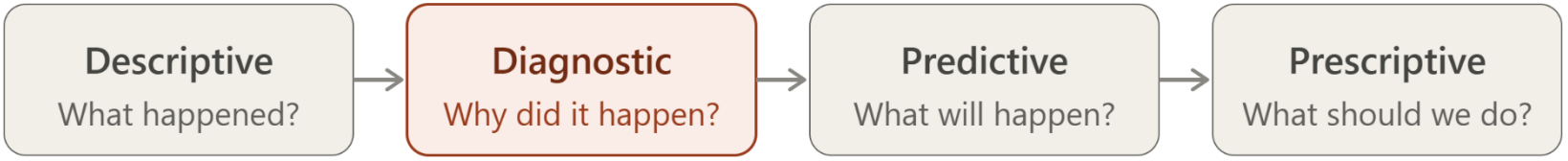
- Diagnostic analytics(Causal analysis)comes after descriptive analytics (which tells you what happened?).
- Diagnostic analytics used to figure out why did it happen? And finds the root cause.



https://www.geeksforgeeks.org/data-analysis/what-is-diagnostic-analytics/

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Diagnostic analytics



Diagnostic analytics investigates the "why" behind an observation

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Types of techniques for diagnostic analytics

1. **Correlation**
2. Regression
3. Drill- down analysis
4. Data mining

Correlations

Example:

" Average students score decreased from 85% to 70% this month"

Apply diagnostic analytics.

- Do scores correlate with attendance? (turns out students who were absent more than three days scored much lower.

Correlation: measure and describe the strength and direction of the relationship between two variables.

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- Strength of the variable indicates how closely two variables are related to each other.
- Direction indicates how one variable changes its value as the value of the other variable changes.
- Correlation measure that examines how two variables change together over time.

Pearson's r correlation

The correlation coefficient (***r***) measures both the strength and direction of a linear relationship between the variables.

Formula:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

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where:

- x_i, y_i= **data values**
- $\bar{x}$, $\bar{y}$= **means of the variables**
- $x_i - \bar{x}$ → **X deviation**
- $y_i - \bar{y}$ → **Y deviation**

Example

Person	X(height)	Y(weight)	$X-X^-$	$Y-Y^-$	$(X-X^-)(Y-Y^-)$	$(X-X^-)2$	$(Y-Y^-)2$
A	150	50	−10	−8.8	88.0	100	77.44
B	155	55	−5	−3.8	19.0	25	14.44
C	160	58	0	−0.8	0.0	0	0.64
D	165	63	5	4.2	21.0	25	17.64
E	170	68	10	9.2	92.0	100	84.64
Total					220	250	194.80

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$$r \approx 0.997$$

Substitute these values in the formula

Interpretations:

Correlation is measured with the **correlation coefficient (r)**, which ranges from **-1 to +1**:

1. Positive correlation: as one variable increases other variable also increases.
2. Negative correlation: one variable increases other variable decreases.
3. No correction: variable has no relationship

Applications of diagnostic analytics

1. Human resource: helps to understand the employee behavior and improve employee retention.
2. Healthcare: to optimize discharge plans and increase operational efficiency.
3. Manufacturing: helps to streamline production process and reduce the cost.
4. Retail: optimize the marketing strategies.

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1. DESCRIPTIVE

What happened?



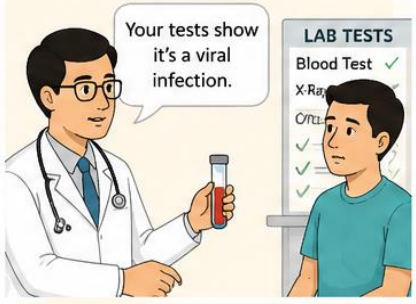
The doctor records and summarizes the patient's symptoms and signs.

Example:
Fever: 102°F, Cough, Headache, Tiredness

Focus:
Understanding what is happening.

2. DIAGNOSTIC

Why did it happen?



The doctor analyzes test results and medical history to find the root cause.

Example:
Viral infection is the cause of the fever.

Focus:
Finding the cause of the problem.

3. PREDICTIVE

What will happen?



The doctor uses data and past patterns to predict the possible outcomes.

Example:
You may recover in 5–7 days with treatment. Without treatment, the condition may worsen.

Focus:
Forecasting future outcomes.

4. PRESCRIPTIVE

What should we do?



The doctor recommends the best actions to achieve the desired outcome.

Example:
Take medicines, rest, drink fluids and come for follow-up.

Focus:
Recommending the best action to achieve the goal.

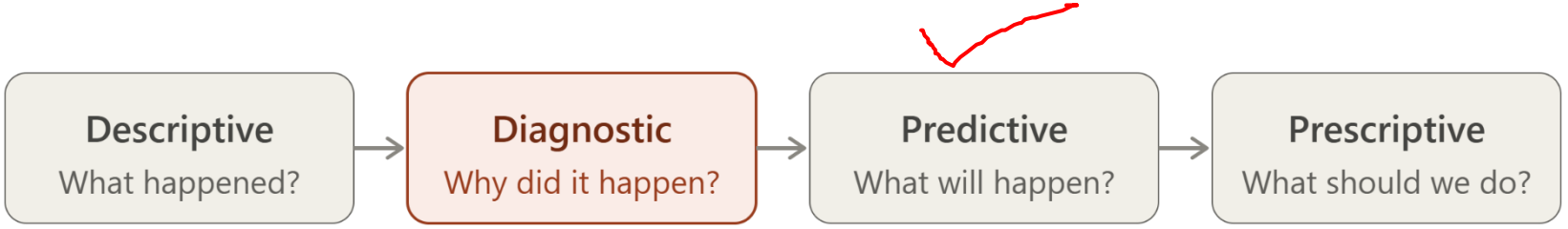
In short: Understand the problem (Descriptive) → Find the cause (Diagnostic) → Predict the future (Predictive) → Take the best action (Prescriptive)

Data + Analysis + Decision = Better Health Outcomes

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3. Predictive analytics



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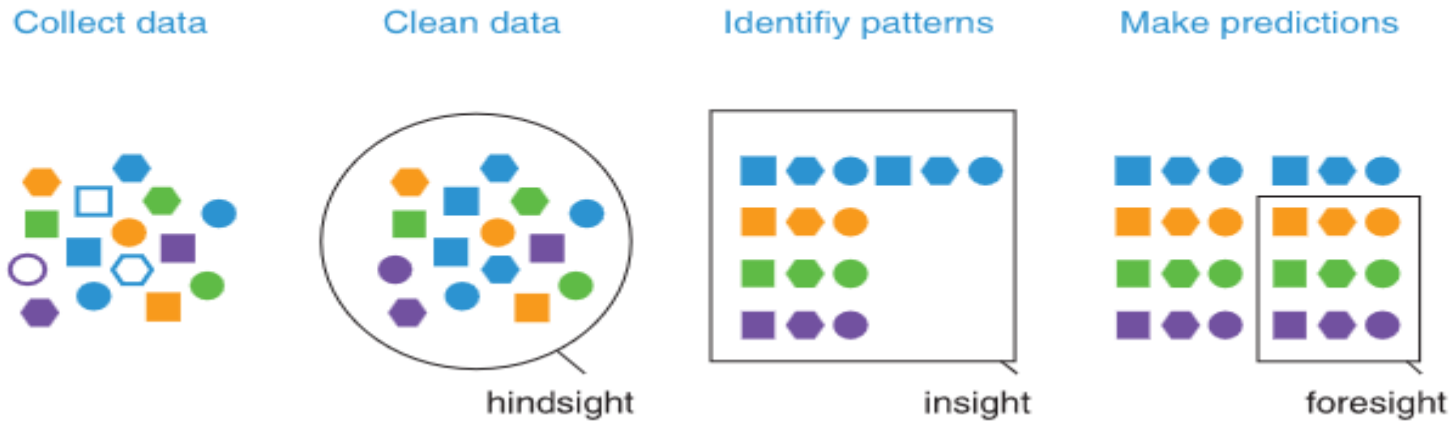
- Predictive analytics answers the questions like **“ what is likely to happen next”.**
- Predictive analytics uses the historical data, patterns(learned through correlation) and machine learning algorithms to predict the future outcomes or trends.
- **Software for predictive analytics:** SAS predictive analytics, IBM predictive analytics, RapidMiner, and others.
- **Techniques:** Linear regression, logistic regression, Decision tree, Random forest, Support vector Machine, Naïve Bayes, K- nearest Neibours.

Process of predictive analytics

Stages in predictive analytics

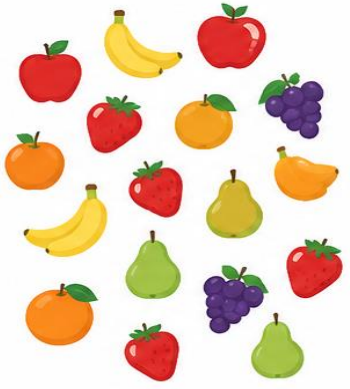
1. Collection of data
2. Cleaning data: (hindsight).
3. Identify patterns: insights
4. Make predictions: Foresight

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Collect Data

Collect raw data about different fruits.



Unorganized and incomplete data

Clean Data

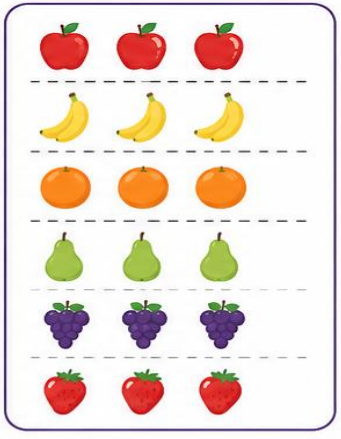
Remove duplicates, fix errors and handle missing values.



Cleaned and consistent data

Identify Patterns

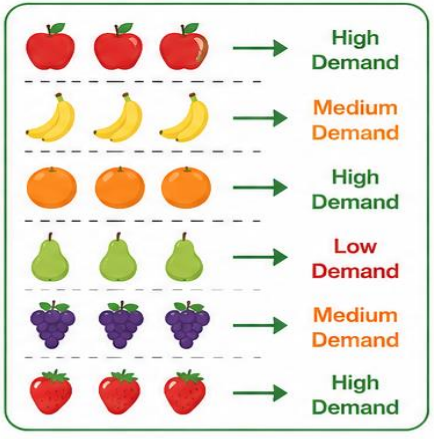
Find patterns and relationships in the clean data.



Patterns / insights are discovered

Make Predictions

Use patterns to predict future outcomes for new fruits data.



Predictions for future demand / sales

Hindsight
Understanding the past data

Insight
Discovering meaningful patterns

Foresight
Predicting the future outcomes

Example: Using past data of fruits (color, type, season, price, sales), we clean the data, find patterns (like apples and oranges sell more in summer), and predict future demand.

4 TYPES OF ANALYTICS – DOCTOR & PATIENT EXAMPLE

From understanding the problem to taking the best action

1. DESCRIPTIVE

What happened?



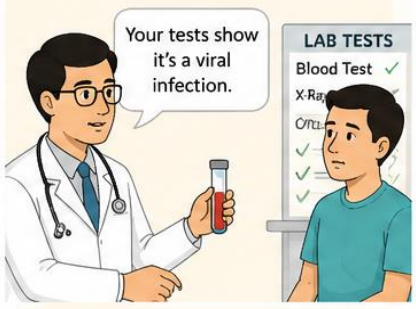
The doctor records and summarizes the patient's symptoms and signs.

Example:
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In short: Understand the problem (Descriptive) → Find the cause (Diagnostic) → Predict the future (Predictive) → Take the best action (Prescriptive)

Data + Analysis + Decision = Better Health Outcomes

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4. Prescriptive analytics

- Prescriptive analytics takes the data from all the previous stages such as what happened, why it happened, what will likely to happen next and tells "what should we do"?
- Prescriptive analytics is area of business analytics.
- prescriptive analytics forecast the future outcomes and also provides best recommendations to achieve best possible outcomes.
- Techniques

1. optimization models 2. simulations 3. game theory 4. decision trees 5. reinforcement learning.

- Example:

- **A hospital predicts that a patient has an 80% chance of developing diabetes in the next five years.**
- **Predictive Analytics:** The patient has an 80% risk of diabetes.
- **Prescriptive Analytics:** The doctor recommends:
  - Exercise 30 minutes daily.
  - Follow a low-sugar diet.
  - Reduce body weight by 5 kg.
  - Schedule regular health checkups.
- The prediction tells **what is likely to happen**, while the prescription tells **what should be done**.

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5. Exploratory analysis

- Exploratory analysis(exploratory data analysis): process of exploring, understanding, cleaning, summarizing, and visualizing a dataset before applying statistical analysis or machine learning algorithms.
- Visual representation of the data.
- Understands about the characteristics of data
- Goals of Exploratory analysis

1. Understand structure of the data: no of rows/ columns, type of data.
2. Find missing data
3. Spot outliers

- Visualization techniques: histogram, pie chart, heatmap, Box plot, scatter plot etc.

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6. Mechanistic analysis

- Mechanistic analysis understands the exact cause and effect relationships between the variables.
- Answers questions like " exactly how does it happen?".
- Studying the relationship between the variables
- **Regression** used to explore the relationship between variables.

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5.1 Regression

- Regression analysis is a statistical method used to estimate the relationship between variables and predict the value of a dependent variable using one or more independent variables.
- Unlike correlation, which only measures the strength and direction of a relationship, regression can predict the value of one variable based on another.
- Regression helps determine how changes in independent variables affect the dependent variable.
- Regression analysis predict outcome variable(dependent variable) from one predictor variable(independent variable). (linear regression)
- Predict outcome from several predictor variable is called multiple linear regression.

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- Linear regression: establishes relationship between the variables
- How does one variable changes when another variable changes.
- EX: how does temperature affect the electricity consumption?

- Linear regression assumes that the relationship between the predictor (independent) variable and the outcome (dependent) variable is linear, meaning it can be represented by a straight line.
- If:

x = Predictor (Independent Variable) y = Outcome (Dependent Variable)then the linear relationship is expressed as:

$$y = \beta_0 + \beta_1 x$$

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- where:
- β_0**(Intercept):** The value of y when $x = 0$. It is the point where the regression line intersects the y-axis.
- β_1**(Slope):** Represents the rate at which y changes for every one-unit increase in x.
- The main objective of **linear regression** is to **estimate the values of** β_0 **and** β_1 using observed data.
- Once these values are estimated, the regression equation can be used to **predict the value of** y for a new value of x.
- The slope (β_1) can also be calculated using the correlation coefficient:
- $\beta_1 = r \times \frac{s_y}{s_x}$
- where:
- r = Pearson correlation coefficient
- s_y = Standard deviation of the dependent variable (y)
- s_x = Standard deviation of the independent variable (x)

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- After calculating the slope (β_1), the **intercept (β_0)** is calculated using:
- $\beta_0 = \bar{y} - \beta_1 \bar{x}$
- where:
- $\bar{x}$= Mean of the independent variable
- $\bar{y}$= Mean of the dependent variable.
- Once both β_0**(intercept)** and β_1**(slope)** are determined, the **linear regression equation**
- $y = \beta_0 + \beta_1 x$
- can be used to **predict the value of the dependent variable (y)** for any given value of the independent variable (x).

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Example

Predict the salary of an employee based on the experience

Employee	Experience (Years) (X)	Salary (₹ Lakhs) (Y)
E1	1	3.0
E2	2	4.0
E3	3	5.0
E4	4	6.0
E5	5	7.0
E6	6	8.0
E7	7	9.0
E8	8	10.0

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Step-1: calculate mean of experience

$$\bar{x} = \frac{1+2+3+4+5+6+7+8}{8} = 4.5$$

Step-2: calculate mean of salary

$$\bar{y} = \frac{3+4+5+6+7+8+9+10}{8} = 6.5$$

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Step-3: calculate standard deviation of x

$$\boxed{s_x = 2.449}$$

Step-4: calculate standard deviation of y

$$\boxed{s_y = 2.449}$$

Step-5: Calculate Pearson Correlation (r)

$$\boxed{r = 1}$$

Indicates positive correlation

Step-6: Calculate slope

$$\beta_1 = r\frac{s_y}{s_x}$$

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$$\beta_1 = 1 \times \frac{2.449}{2.449}$$

$$= 1 \times 1$$

$$\boxed{\beta_1 = 1}$$

For every one unit increase in experience salary increased by 1 lakh

Step-7: Calculate the intercept

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

- Substitute the values
- $= 6.5 - (1 \times 4.5) = 2$
- Therefore,

$$\boxed{\beta_0 = 2}$$

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Interpretation

- When experience is 0 years, the predicted starting salary is ₹2 lakhs.

If new employee has 10 years experience what is his predicted salary

X=10

Salary=2+1*10=12L

Absentees: 5/8/2026- 144,146,151,155,158,159,65,67,190,91,99.

Statistics: Understanding attributes and their types, Types of attributes, Discrete and continuous attributes, measuring central tendency: Mean, Mode, Median, measuring dispersion, Skewness and kurtosis, understanding relationships using covariance and correlation coefficients: Pearson's correlation coefficient, Spearman's rank correlation coefficient, collecting samples, performing parametric tests.

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Statistics

- Before preprocessing and analysis we must familiar with the data here statistics helps to get initial insights and understanding of the data.
- Example: arithmetic mean of the employee working hours understands the load of an employee in an organization.
- Correlation helps us to understand the relationship between the two variables.

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Understanding attributes and their types

- Attribute: column or data field or series that represents the characteristics of an object.
- Variable, a feature or dimension.
- Statisticians call it as variable.
- Machine learning engineer- feature
- Data warehouse- dimension
- Database professionals- attribute.

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Types of attributes

- Before analyzing the data we must understand what type of attributes are there in the dataset.
- Helps us to choose correct method of data analysis and visualization techniques.
- List of attributes

1. Nominal attributes
2. Ordinal attributes
3. Numerical attributes

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Types of attributes..

1. **Nominal attributes:** names or labels used to classify the data into different categories.

   - qualitative and no order in nature

   - ex: gender, colors, product names, marital status, blood group.

   - as they are categorical values, mean and median cant be applied.

   - mode is applied on categorical data to identify the most frequently occurred category.

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2. Ordinal attributes: categorical (qualitative) data with a meaningful order or ranking, but the difference between the ranks cannot be measured.

- used to represent opinions, preferences, or sati factionary levels.

- examples: movie rating, customer satisfaction- very dissatisfied, satisfied, neutral, very satisfied.
- Attributes can be measured with mode and median.

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NOMINAL ATTRIBUTES

(Categories / Labels – No Natural Order)

Nominal attributes are names or labels
used to categorize data.
There is no inherent order or ranking.

Examples

1. Gender



2. Blood Group



3. Colors



4. Vehicle Type



5. Fruit Type



6. Country



▶ **Key Point:** Nominal data only represent categories.
There is no order, so we cannot rank them.

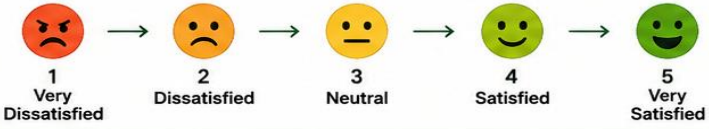
ORDINAL ATTRIBUTES

(Categories with a Natural Order / Ranking)

Ordinal attributes have a meaningful order or ranking,
but the exact difference between the ranks
is not known.

Examples

1. Customer Satisfaction Rating



2. Product Quality Rating



3. Education Level



4. Movie Rating (Example)



▶ **Key Point:** Ordinal data have a clear order or ranking,
but the exact difference between ranks is not measurable.

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3. Numerical attributes

- numeric attributes represented as integers or real numbers.

- classified into two types-1. interval scaled attributes 2. ratio scaled attributes

1. interval scaled attributes

- meaningful order and equal intervals between them

- there is no true zero. (e.g., 0°C does not mean there is no temperature).

- addition and subtraction can be applied

- multiplication and division are not meaningful.

- mean, mode and median can be calculated.

- examples: temperature, calender year.

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2. ratio scaled attributes

- Values have a meaningful order, equal intervals, and a true (absolute) zero.

- Addition, subtraction, multiplication, and division can all be performed.
- The mean, median, and mode can be calculated.
- Examples: Height, weight, latitude, longitude, years of experience, number of words in a document.
- Kelvin temperature is a ratio scale because it has a true zero, whereas Celsius and Fahrenheit are interval scales.

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Discrete and continues attributes

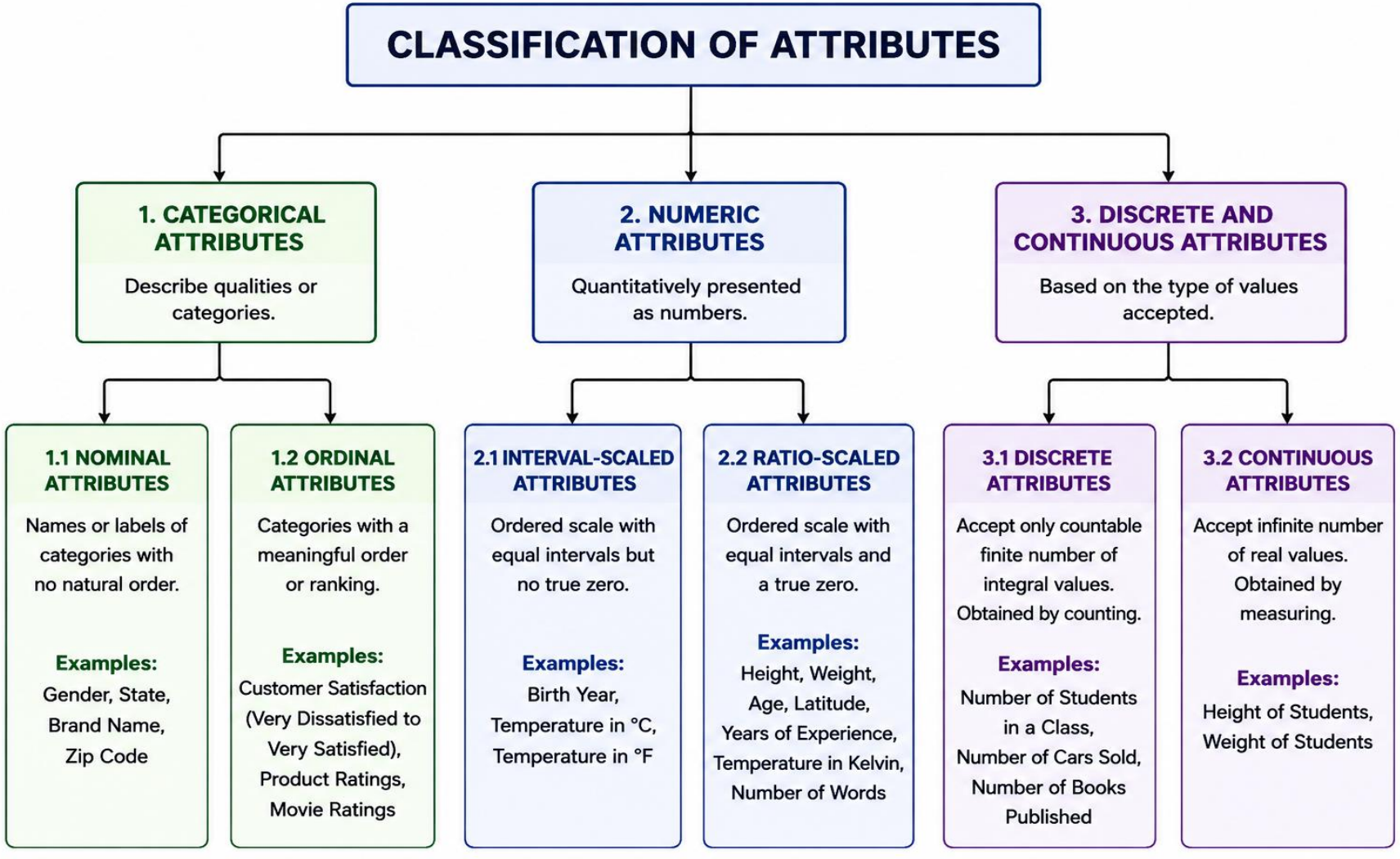
Attributes can also be classified as discrete or continues attributes.

1. Discrete attribute
   - contains countable finite number of values.
   - No of students in a class, total no of cars sold.
   - Accepts only whole numbers.

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2. Continues attribute
   - accepts infinite number of possible values.
   - Accepts real values.
   - Fractions make sense
   - Height and weight of students.

Summary of classification of attributes



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Statistics: Understanding attributes and their types, Types of attributes, Discrete and continuous attributes, measuring central tendency: Mean, Mode, Median, measuring dispersion, Skewness and kurtosis, understanding relationships using covariance and correlation coefficients: Pearson's correlation coefficient, Spearman's rank correlation

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Measuring central tendency

- Central tendency refers to the tendency of data values to cluster around a central or average value.
- Helps to represent entire dataset with single value.
- The three common measures of central tendency.

1. Mean
2. Median
3. mode

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1. Mean

- The mean is the arithmetic average of a set of observations. It is calculated by adding all the observation values and dividing the sum by the total number of observations.
- The mean is sensitive to outliers and noise, meaning that unusually high or low values can significantly affect the average, causing it to deviate from the typical central value.

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$$\text{Mean} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

- where:
- x_i = each observation
- N = total number of observations

Example

Name	Gender	Communication Skill Score	Quantitative Skill Score
John	M	40	38
Alia	F	45	41
Ananya	F	23	42
Steve	M	39	48
Ben	M	39	32

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Calculate mean of the communication skill score using pandas library

```
# Import pandas library
import pandas as pd

# Create dataframe
sample_data = {'name': ['John', 'Alia', 'Ananya', 'Steve', 'Ben'],
               'gender': ['M', 'F', 'F', 'M', 'M'],
               'communcation_skill_score': [40, 45, 23, 39, 39],
               'quantitative_skill_score': [38, 41, 42, 48, 32]}

data = pd.DataFrame(sample_data, columns = ['name', 'gender', 'communcation_skill_score', 'quantitative_skill_score'])
data.head()
# find mean of communication_skill_score column
data['communcation_skill_score'].mean(axis=0)
```

Out[1]:

	name	gender	communcation_skill_score	quantitative_skill_score
0	John	M	40	38
1	Alia	F	45	41
2	Ananya	F	23	42
3	Steve	M	39	48
4	Ben	M	39	32

np.float64(37.2)

2. Mode

- Mostly used for the categorical values.
- If all the values in the group are unique or no repeated then no mode.

```
# find mode of communication_skill_score column
data['communcation_skill_score'].mode()[0]
```

- Output - np.int64(39)

3. Median

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- Median is a middle value in a group of observations.
- Median is less affected by outliers compared with the mean.
- Therefore, it is often a better measure of central tendency when data contains extreme values.
- In Pandas, use the median() function to calculate the median.

```
# find median of communication_skill_score column
data['communcation_skill_score'].median()
```

```
39.0
```

Measuring dispersion

- Measures the deviation in the data values of dataset.
- Most popular dispersion techniques are

1. Range 2. interquartile range 3. variance 4. standard deviation

1. Range:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

```
column_range=data['communcation_skill_score'].max()-data['communcation_skill_score'].min()
print(column_range)
```

```
22
```

2. Interquartile range:

```
# Calculate Interquartile Range
#First Quartile
q1 = data['communcation_skill_score'].quantile(.25)

# Third Quartile
q3 = data['communcation_skill_score'].quantile(.75)

# Inter Quartile Ratio
iqr=q3-q1
print(iqr)
```

```
1.0
```

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3. Variance: measure deviation from the mean.

$$Variance = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

```
# Variance of communication_skill_score
data['communcation_skill_score'].var()
```

```
69.2
```

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4. Standard deviation: square root of variance

- Its unit is the same as the original data, making it easy to interpret.
- Low SD → observations are close to the mean → less spread.
- High SD → observations are far from the mean → more spread.

```
# Standard deviation of communication_skill_score
data['communcation_skill_score'].std()
```

```
8.318653737234168
```

- **describe()** metho is used to get the summary of all the statistics in a single command.
- Returns count, mean, standard deviation, first quartile, median, third quartile, minimum and maximum values of each numeric columns.

```
# Describe dataframe
print(data.describe())
```

```
...       communcation_skill_score  quantitative_skill_score
count                     5.000000                  5.000000
mean                     37.200000                 40.200000
std                       8.318654                  5.848077
min                      23.000000                 32.000000
25%                      39.000000                 38.000000
50%                      39.000000                 41.000000
75%                      40.000000                 42.000000
max                      45.000000                 48.000000
```

Skewness and kurtosis

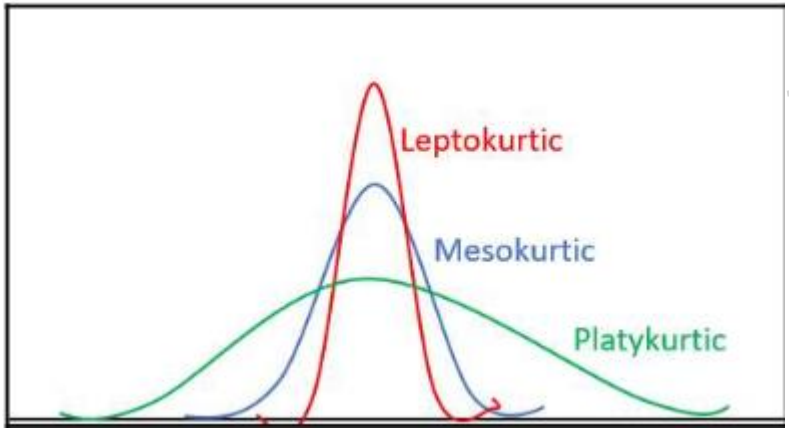
- Skewness measures the symmetry of distribution of data points.
- Shows the how much distribution is deviated from a normal distribution.
- Skewness can be:
  - 0 → approximately symmetric/normal distribution
  - Positive → right-skewed; tail extends toward the right
  - Negative → left-skewed; tail extends toward the left
- Relationship with Mean and Median
  - Positive skewness: Mean > Median > Mode
  - Negative skewness: Mean < Median < Mode

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```
# skewness of communication_skill_score column
data['communcation_skill_score'].skew()
```

```
np.float64(-1.704679180800373)
```

- Kurtosis measures the tailed ness of a distribution compared with a normal distribution.
- It indicates the presence of extreme values (outliers) in the data.
- High kurtosis(Leptokurtic) → heavier tails → more outliers.
- Low kurtosis (Mesokurtic)→ lighter/thinner tails → fewer outliers.



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```
# kurtosis of communication_skill_score column
data['communcation_skill_score'].kurtosis()
```

```
np.float64(3.6010641852384015)
```

Understanding relationships using covariance and correlation coefficients

- Relationship between the variable helps in data analysis to understand the dynamics between variables.
- Covariance measures the relationship between two variables.
- It tells us how two variables change together.
- Example: relationship between communication skill score and quantitative skill score.
- It indicates whether changes in one variable are associated with changes in the other.
- Covariance can have values from $-\infty$ to $+\infty$.
- A limitation is that covariance is not normalized, so its magnitude is difficult to interpret and compare across different datasets.

Covariance between columns of dataframe

- data.cov()

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Pearson's correlation coefficient

- Correlation shows how one variable correlate with other variable.
- Refer slide nos : 41 to 43.

```
# Correlation between columns of dataframe
data.corr(method ='pearson', numeric_only=True)
```

	communcation_skill_score	quantitative_skill_score
communcation_skill_score	1.00000	-0.13464
quantitative_skill_score	-0.13464	1.00000

- in above code 'numeric_only' applies only for the numeric columns.
- attribute method has following values.
  - pearson: standard correclation coefficient
  - Kendall: Kendell's tau correlation coefficient
  - Spearman: Spearman's rank correlation coefficient.

Spearman's rank correlation coefficient

- Spearman's correlation measures the strength and direction of the relationship between two variables based on their ranks.
- It is a non-parametric correlation method.
- First, the observations are converted into ranks, and then correlation is calculated between the ranks.
- It is suitable for ordinal data and can also be used for continuous and discrete data.
- It is useful when:
  - Data is not normally distributed
  - Data contains outliers
  - The relationship is monotonic rather than necessarily linear
- Spearman's correlation coefficient ranges from −1 to +1.

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Types of samplings

- probability sampling
- non probability sampling.

Parametric tests

T- test

- one way
- two way
- paired t-test

ANOVA

- one way ANOVA

two way ANOVA

N- way multi variant ANOVA

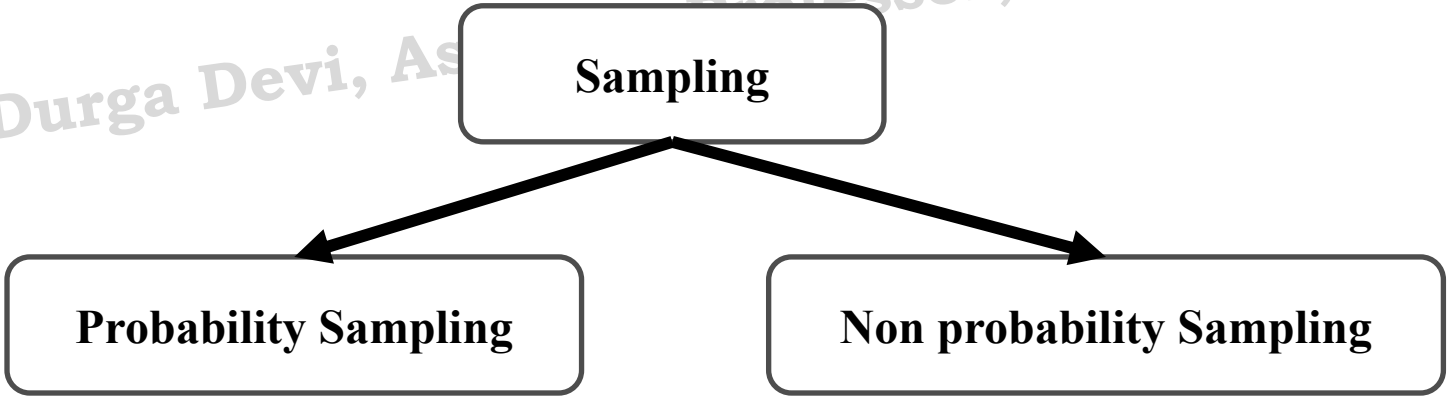
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Collecting samples

- Sample is a small subset of population used for the data analysis.
- Sampling: the process of selecting sample from the given population.
- Types of sampling techniques

1. Probability sampling
2. Non- probability sampling

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1. **Probability sampling:** members of the population selected randomly.

each member of population is equal chance of selecting in a sample.

following are techniques of probability sampling

a) simple random sampling

b) Stratified sampling

c) systematic sampling

d) cluster sampling

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Probabilistic sampling

a) Simple random sampling:

- ✓ every member of a sample has an equal chances of being selected
- ✓ Example: 500 students randomly selected from 5000 students for survey.

b) Stratified sampling:

- ✓ Divides the entire population into smaller groups known as strata.
- ✓ select the sample from each stratum.
- ✓ Each stratum is not equal in size.
- ✓ Improves the accuracy and reduces the selection bias.
- ✓ Example: college has 5000 students divided them into branches. Such as CSE, CSE-AIML, AIDS, IT etc. from each branch select the students randomly.

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Probabilistic sampling

c) Systematic sampling:

- ✓ select members at regular intervals.
- ✓ members are selected in systematic order such as every nth member from the population.
- ✓ From a list of students in each section, randomly choose a starting student and then select every 5th student for feedback.

d) Cluster sampling:

- ✓ **divide the entire population into clusters(groups)**
- ✓ **clusters are formed based on the gender, location, occupation and so on.**
- ✓ **example :** Suppose a university has 20 departments. Instead of selecting individual students from all 20 departments. Randomly select 5 departments, and survey students from those selected departments.

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2. Nonprobability sampling

- Non-probability sampling means that participants are not selected randomly, so every person does not have a known or equal chance of being selected.
- Cheaper, faster and convenient
- can be biased.

Types of non probability sampling

1. Convenience sampling
2. Purposive sampling
3. Quota sampling
4. Snowball sampling

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Non probability sampling

a) Convenience sampling :

- select members based on the availability and willingness to participate.
- easier and convenient method.
- Usually cheaper and can be biased.
- Example: survey the first 100 students you meet in the college canteen.

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b) Purposive sampling

- also called **judgmental sampling.**
- Selection is based on the researcher judgement.
- Researcher intentionally select the member who have knowledge relevant to the study.
- Example: a study on AI teaching practices, specially select only CSE faculty who teaches AI/ML courses.

Non probability sampling

c) Quota sampling :

- Determine how many members are needed from different groups.
- Members are selected until it reaches the required quota.
- Selection process need not be random.
- Example: you want 100 students. 50 students from male group and 50 from female.

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d) Snowball sampling

- Used when finding respondents are rare and difficult to trace for study.
- Researcher starts with few known respondents
- Respondents refer other respondents.
- Example : finding free lancers for the survey. Research starts survey with two free lancers and they refer other free lancers.

- Types of sampling techniques

1. probability- simple random, systematic, stratified, cluster

2. non probability- convenient, quota, purposive, snowball samplings

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Perform parametric tests

- Inferential statistics are one of the branch of statistics used to draw the conclusion from the sample data about the larger population.
- Hypothesis testing is one of the part of the inferential statistics.
- Hypothesis testing is a systematic method for deciding whether the evidence from a sample is strong enough to support a claim about a population.
- Example: "We do not know the actual average of the entire population. The college has made a claim that the average is 60. We take a sample from the population, and the sample average is 65. Now we need to determine whether there is sufficient statistical evidence to say that the population average is actually different from 60."
- Compare the sample mean with the hypothesized mean. Use parametric test.
- Parametric tests are used for quantitative and continuous data.

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Types of parametric tests

1. **T-test** is a parametric test used to determine whether there is a significant difference between means.

- It is commonly used when the data approximately follows a normal distribution.
- Two common types are:
  - One-sample t-test – compares a sample mean with a hypothesized population mean.
  - Two-sample t-test – compares the means of two groups.
- Example: Check whether the average weight of 10 students is significantly different from 68 kg using a one-sample t-test.

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One sample t-test

```
import numpy as np

from scipy.stats import ttest_1samp

data=np.array([63, 75, 84, 58, 52, 96, 63, 55, 76, 83])

mean_value = np.mean(data)

print("Mean:",mean_value)

t_test_value, p_value = ttest_1samp(data, 68)

print("P Value:",p_value)

print("t-test Value:",t_test_value)

# 0.05 or 5% is significance level or alpha.

if p_value < 0.05:

    print("Hypothesis Rejected")

else:

    print("Hypothesis Accepted")
```

ttest_1samp () returns
T-statistics and p- value
P- value used to evidence to prove our assumptions.

```
Mean: 70.5
P Value: 0.5986851106160134
t-test Value: 0.545472577903943
Hypothesis Accepted
```

Output shows hypothesis is accepted with 95% confidence level means average weight of the student of a class is 68kgs.

Two-sample t-test(independent sample test)

Two- sample t-test used to compare the mean of two independent groups.

Example: compare average height of two student groups

1. Null Hypothesis H_0 : H_0:$\mu 1=\mu 2$
2. **Alternative Hypothesis H_a:** H_a:μ_1 not equal μ_2. $\mu_1 > \mu_2$ or $\mu_2 > \mu_1$?

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2-sample t-test

```
from scipy.stats import ttest_ind
# ttest_ind function for two independent samples

data1=np.array([63, 75, 84, 58, 52, 96, 63, 55, 76, 83])

data2=np.array([53, 43, 31, 113, 33, 57, 27, 23, 24, 43])

# Compare samples
stat, p = ttest_ind(data1, data2)

print("p-values:",p)

print("t-test:",stat)
# 0.05 or 5% is significance level or alpha.

if p  < 0.05:

    print("Hypothesis Rejected")

else:

    print("Hypothesis Accepted")
```

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T- statistics: tells us how large the difference between the two sample means.

P- value: It helps us decide whether the observed difference between the two means is statistically significant.

```
p-values: 0.015170931362451 26
t-test: 2.683587991381918
Hypothesis Rejected
```

Null hypothesis is rejected with 95% confidence. Sample Means are different.

Paired t- test

- A paired sample t-test is used when we have two measurements that are related to each other.
- Here also performing two- sample t-set.
- "Before vs After measurement of the same people".
- Example: compare the blood pressure levels of same people before and after the drug treatment.

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- Example: take weights of the people before and after their weight loss treatment.

1. Null Hypothesis H_0 : H_0:$\mu 1=\mu 2$- mean difference between groups is zero.
2. Alternative Hypothesis H_a: H_a:μ_1 not equal μ_2. $\mu_1 > \mu_2$ or $\mu_2 > \mu_1$?- mean difference between two sample groups are not same.

Example on paired t- test

```
# paired test
from scipy.stats import ttest_rel

# Weights before treatment
data1=np.array([63, 75, 84, 58, 52, 96, 63, 65, 76, 83])

# Weights after treatment
data2=np.array([53, 43, 67, 59, 48, 57, 65, 58, 64, 72])

# Compare weights
stat, p = ttest_rel(data1, data2)

print("p-values:",p)

print("t-test:",stat)
# 0.05 or 5% is the significance level or alpha.

if p  < 0.05:

    print("Hypothesis Rejected")

else:

    print("Hypothesis Accepted")
```

```
p-values: 0.013685575312467715
t-test: 3.0548295044306903
Hypothesis Rejected
```

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Interpretation of results:

- Null hypothesis is rejected with 95% confidence
- Weight loss treatment has a significant impact

On the patient weight.

ANOVA

- ANOVA(ANalysis of VAriance): one of the statistical inference test used to compare multiple groups at a time.
- Compares the more than two sets of data and checks statistical significance.
- Example: compare average employee performance of three different groups.(different locations).

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ANOVA essentially compares:

$$\frac{\text{Variation between groups}}{\text{Variation within groups}}$$

- This ratio is called F- statistics.

- ANOVA three ways

1. one- way ANOVA
2. Two- way ANOVA
3. N-way multivariate ANOVA

One- way ANOVA method

1. Null Hypothesis H_0 : there is no difference between mean performance score of multiple locations.
2. Alternative Hypothesis H_a: there is a difference between mean performance score of multiple locations.

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One-way ANOVA

```
from scipy.stats import f_oneway
# Performance scores of Mumbai location
mumbai=[0.14730927, 0.59168541, 0.85677052, 0.27315387, 0.78591207,0.52426114, 0.05007655, 0.64405363, 0.9825853 , 0.62667439]
# Performance scores of Chicago location
chicago=[0.99140754, 0.76960782, 0.51370154, 0.85041028, 0.19485391,0.25269917, 0.19925735, 0.80048387, 0.98381235, 0.5864963 ]
# Performance scores of London location
london=[0.40382226, 0.51613408, 0.39374473, 0.0689976 , 0.28035865,0.56326686, 0.66735357, 0.06786065, 0.21013306, 0.86503358]
# Compare results using Oneway ANOVA
stat, p = f_oneway(mumbai, chicago, london)
print("p-values:", p)
print("t-test:", stat)
if p  < 0.05:
    print("Hypothesis Rejected")
else:

    print("Hypothesis Accepted")
```

```
p-values: 0.27667556390705783
t-test: 1.348044638196545
Hypothesis Accepted
```

Interpretation of results:

- There is no significant difference between performance of employees score.

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2. two- way ANOVA: comparing multiple groups based on two independent variables.

Example: IT company wants to compare employee productivity based on the employees working hours and project complexity

- Depenedent variable: employee productivity
- Independent variable: working hours
- Independent variable: project complexity

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3. N-way multivariant ANOVA: compare multiple groups based on N-independent variables.

Example : IT company wants to study employee productivity based on

1. Working hours
2. Project complexity
3. Employee training
4. Workplace facilities.

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References

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